

USER MANUAL

Certificate System DApp

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1. Introduction

1.1 Purpose of the System

The Certificate System DApp is a decentralized application that allows academic institutions to issue tamper-proof digital certificates. Employers can instantly verify these certificates online without contacting the institution.

1.2 Who Should Use This

User Type	Description
Lecturers	Issue certificates to students
Students	Receive and view certificates
Employers	Verify certificate authenticity
Admins	Manage lecturers and system settings

2. Prerequisites

2.1 Installing MetaMask

- Go to: <https://metamask.io/download/>
- Click "Download"
- Install the extension for your browser: Chrome / Brave / Edge
- Create a wallet or import an existing one

2.2 Adding Sepolia Network

- Click the MetaMask extension icon
- Click the network dropdown (top left)
- Click "Add Network"
- Enter these details:

Field	Value
Network Name	Sepolia
RPC URL	https://rpc.sepolia.org
Chain ID	11155111
Currency Symbol	SepoliaETH
Block Explorer	https://sepolia.etherscan.io

Click "Save"

2.3 Getting Sepolia ETH (For Testing)

- Go to: <https://sepolia-faucet.pk910.de/>

- Paste your wallet address
- Complete the captcha
- Click "Send me ETH"
- Wait for the transaction to confirm

3. Getting Started

3.1 Opening the DApp

- Open your browser
- Navigate to: <http://127.0.0.1:5500/frontend/index.html>
- Or open the index.html file directly

3.2 Connecting Your Wallet

Click "Connect Wallet" button (top right)

```
CertificateChain
Home Issue Verify Dashboard Events
• Sepolia Disconnected [Connect Wallet]
```

MetaMask will pop up:

```
MetaMask

Connect this website with MetaMask

• Account 1 0xBc89D...
○ Account 2 0x41Cde...

[Cancel] [Connect]
```

- Select your account
- Click "Connect"

3.3 After Connecting

```
CertificateChain
Home Issue Verify Dashboard Events
    • Sepolia Connected 0xbc89...8039
      [Connected] [v]
```

4. User Roles

4.1 Checking Your Role

- Go to the Dashboard page
- Look at the "Role" badge

Role	Badge Color	What You Can Do
Lecturer	Blue	Issue certificates
Student	Gray	View certificates
User	Gray	Verify certificates

4.2 Role Capabilities

Action	Lecturer	Student	User
Issue Certificate	Yes	No	No
Verify Certificate	Yes	Yes	Yes
View Dashboard	Yes	Yes	Yes
View Events	Yes	Yes	Yes
View Certificates	Yes	Yes	No

5. How to Issue a Certificate

5.1 Prerequisites

- You are connected as a Lecturer
- You have the student's wallet address

- You have a certificate document uploaded to IPFS

5.2 Step-by-Step

Step 1: Go to "Issue" Page

Click the "Issue" tab in the navigation

Step 2: Fill in Student Details

Field	Description	Example
Student Address	Student's wallet address	0x123456...
Student Name	Student's full name	Jerry Dorlopi
Course Name	Course/program name	Blockchain Development
Grade	Grade achieved	A+

Step 3: Enter IPFS Hash

Upload your certificate document to IPFS (using Pinata). Copy the CID and paste it in the "IPFS Hash" field.

Step 4: Click "Issue Certificate"

Click the "Issue Certificate" button. MetaMask will pop up.

Step 5: Confirm Transaction

Click "Confirm" in MetaMask. Wait for the transaction to be mined (10-30 seconds).

Step 6: Success!

```
Certificate issued successfully!
Token ID: 2
Transaction: 0x5e6be3285af08176467a3ef98c0bb7523d554915...
```

6. How to Verify a Certificate

6.1 Verify by Certificate ID

Step 1: Go to "Verify" Page

Click the "Verify" tab.

Step 2: Enter Certificate ID

```
Verify Certificate

Certificate ID (Token ID): [2]

[Verify]
```

Step 3: Click "Verify"

The system will display:

```
Certificate Details

Certificate ID: 2
Student Name: Jerry Dorlophia
Course: Blockchain and Decentralized Apps
Grade: A+
Issue Date: 7/10/2026
IPFS Hash: bafybeienyozqiv6qid6...
Status: Valid
Owner: 0xBc89D1a74c74fD84e2D33B9430E58bd14CC28039

[Certificate Document]

[Certificate Image Preview]
```

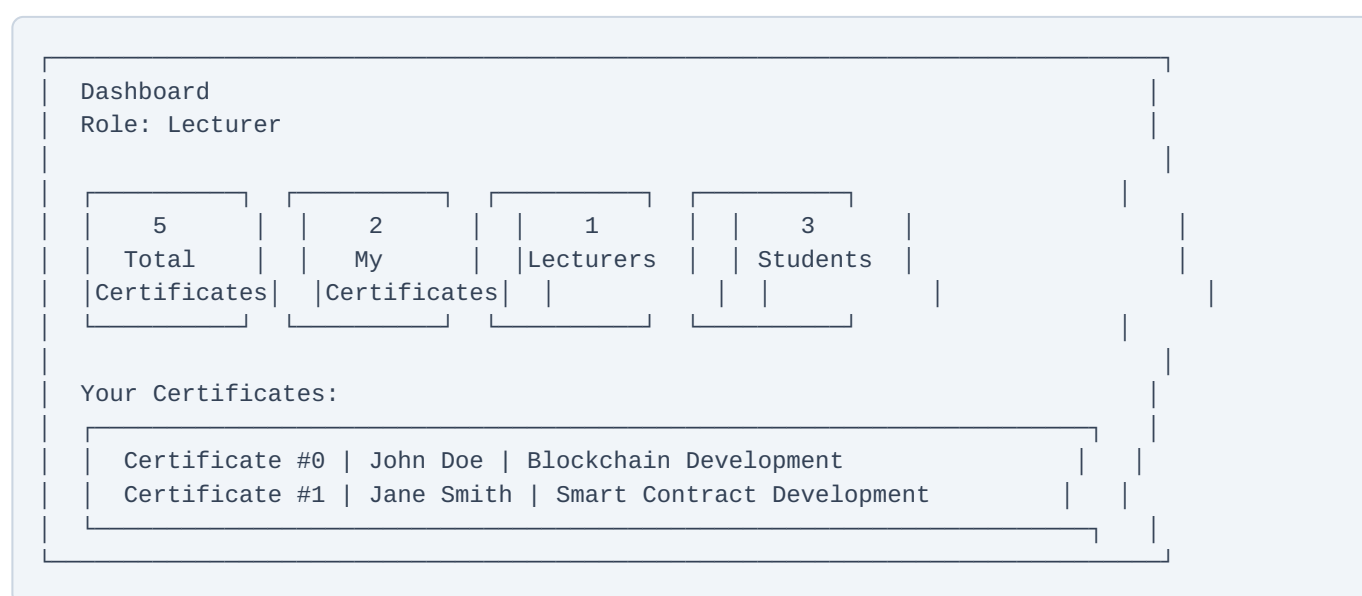
6.2 Verify by QR Code

- Click "Start Scanner"
- Allow camera access
- Point the camera at the QR code
- The certificate will verify automatically

7. How to View Dashboard

7.1 Dashboard Overview

Go to the "Dashboard" page:



7.2 Understanding the Stats

Stat	Description
Total Certificates	All certificates issued in the system
My Certificates	Certificates owned by you
Lecturers	Total number of lecturers
Students	Total number of registered students

8. How to View Blockchain Events

8.1 Events Overview

Go to the "Events" page:



8.2 Event Types

Event	Description
CertificateIssued	A new certificate was issued
CertificateVerified	A certificate was verified
CertificateRevoked	A certificate was revoked
StudentRegistered	A new student registered
LecturerAdded	A new lecturer was added

8.3 Filtering Events

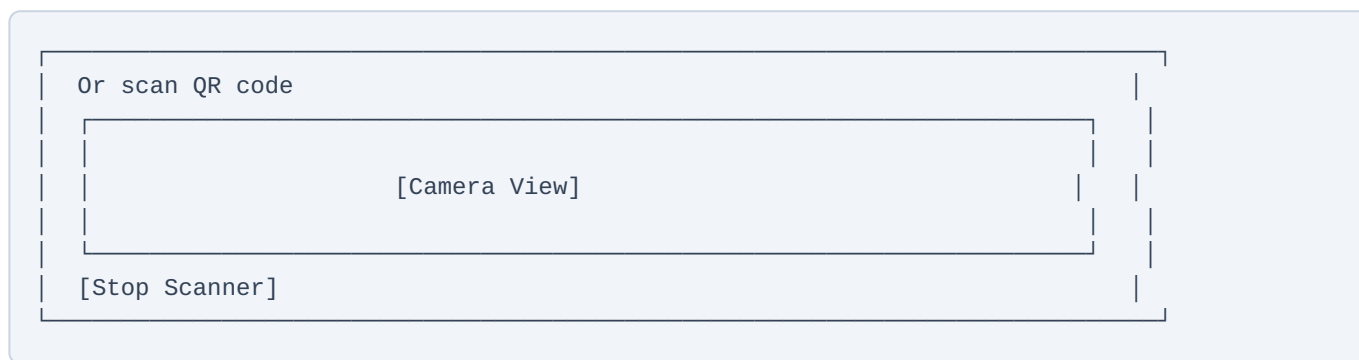
- Use the dropdown to filter by event type
- Click "Refresh Events" to fetch the latest events
- Click "Clear" to clear the display

9. How to Use QR Code Scanner

9.1 Scanning a QR Code

Step 1: Go to "Verify" Page

Step 2: Click "Start Scanner"



Step 3: Allow Camera Access

Click "Allow" when your browser asks for camera permission.

Step 4: Scan the QR Code

Point the camera at the QR code. The certificate ID will be detected automatically and details will appear.

10. Troubleshooting

10.1 Common Issues

Issue	Solution
MetaMask not connecting	Refresh page, click "Connect Wallet" again
Insufficient funds	Get Sepolia ETH from a faucet
Transaction failing	Check you're on Sepolia network
Issue tab hidden	You're not a lecturer
QR scanner not working	Allow camera permissions
Events not loading	Click "Refresh Events"
Certificate image not loading	Check IPFS gateway

10.2 Error Messages

Error	Meaning	Solution
"Only lecturer"	You're not a lecturer	Switch to lecturer account
"Student not registered"	Student isn't registered	Register student first
"Certificate does not exist"	Invalid token ID	Check the ID and try again
"Insufficient contract balance"	Contract has no ETH	Fund the contract

11. FAQ

Q1: Who can issue certificates?

A: Only registered lecturers can issue certificates. The system uses Role-Based Access Control (RBAC) to enforce this.

Q2: Can certificates be revoked?

A: Yes, lecturers can revoke certificates. Once revoked, the certificate shows as "Revoked" when verified.

Q3: Where are certificates stored?

A: Certificate data is stored on the blockchain. Document files are stored on IPFS (decentralized storage).

Q4: Can anyone verify certificates?

A: Yes! Verification is public. Anyone with the certificate ID can verify authenticity.

Q5: Is this system secure?

A: Yes. Blockchain provides immutability. IPFS provides decentralized storage. RBAC ensures only authorized users can issue certificates.

Q6: What is an NFT certificate?

A: Each certificate is an ERC-721 NFT (Non-Fungible Token). This means it's unique, verifiable, and owned by the student.

Q7: How do I get Sepolia ETH?

A: Use a faucet like: <https://sepolia-faucet.pk910.de/>

Q8: What if I forget my certificate ID?

A: Check your Dashboard to see all your certificates and their IDs.

Contact Support

For additional support:

- GitHub: <https://github.com/JerryDorlopia/blockchain>
- Email: [Your Email Address]

Version History

Version	Date	Changes
1.0	July 2026	Initial release

Thank you for using the Certificate System DApp!

How to Add to Your Repository

Create the Docs Folder and Files

```
# Create docs folder
mkdir docs

# Create user-manual.md
# Copy the user manual content above into docs/user-manual.md

# Create architecture-diagram.png
# Generate the diagram using Draw.io or Lucidchart

# Commit and push
git add docs/
git commit -m "Add user manual and system architecture diagram"
git push
```